The documentation and process conversion measures necessary to comply with this revision shall be completed by 17 April 2015.

INCH–POUND

MIL–PRF–19500/415B
14 January 2015
SUPERSEDING
MIL–S–19500/415A
31 October 2003

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, HIGH POWER, STUD MOUNT CASE, TYPES 2N2812 AND 2N2814, QUALITY LEVELS JAN, JANTX, AND JANTXV

Inactive for new design after 7 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL–PRF–19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, high-power transistors. Three levels of product assurance (JAN, JANTX, and JANTXV) are provided for each encapsulated device type as specified in MIL–PRF–19500.

1.2 Package outlines. The device package outline is a TO–210AC (formerly TO–61) in accordance with figure 1 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T (1) T_A = +25°C	P_T (2) T_C = +100°C	V_{CBO}	V_{EBO}	V_{CEO}	I_C	I_B	$R_{\theta JC}$	T_{STG} and T_{OP}
	W	W	V dc	V dc	V dc	A dc	A dc	°C/W	°C
2N2812	4.0	50	80	8.0	60	10	2.0	2.0	–65 to +200
2N2814	4.0	50	120	8.0	80	10	2.0	2.0	

(1) Derate linearly 22.8 mW/°C for T_A > +25°C.
(2) Derate linearly 0.5 mW/°C for T_C > +100°C.

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1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25ºC.

Limit	$V_{BE(sat)}$ I_C = 5.0 A dc I_B = 0.5 A dc	$V_{CE(sat)}$ I_C = 5.0 A dc I_B = 0.5 A dc	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	h_{FE2} V_{CE} = 5.0 V dc I_C = 5.0 A dc	$\|h_{fe}\|$ V_{CE} = 10 V dc I_C = 1.0 A dc f = 10 MHz
	V dc	V dc	pF		
Minimum				40	1.5
Maximum	1.2	0.5	350	120	7

Limit	h_{FE} at V_{CE} = 5 V dc (1)			Switching time		
	h_{FE1} I_C = 1.0 A dc	h_{FE3} I_C = 10 A dc	h_{FE4} I_C = 10 mA dc	t_{ON}	t_s	t_f
				ns	ns	ns
Minimum	50	15	10			
Maximum	150			400	1000	400

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL–PRF–19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet are "JAN", "TX", and "TXV".

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "2812" and "2814".

1.5.3 Suffix symbols. Suffix symbols are not applicable for this specification sheet.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML–19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL–PRF–19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL–STD–750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL–PRF–19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL–PRF–19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL–PRF–19500 and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL–PRF–19500, MIL–STD–750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Pin-out. The pin-out of the device types shall be as shown on figure 1. Terminal 1 is the emitter, terminal 2 is the base, and terminal 3 is the collector. The collector shall be electrically connected to the case.

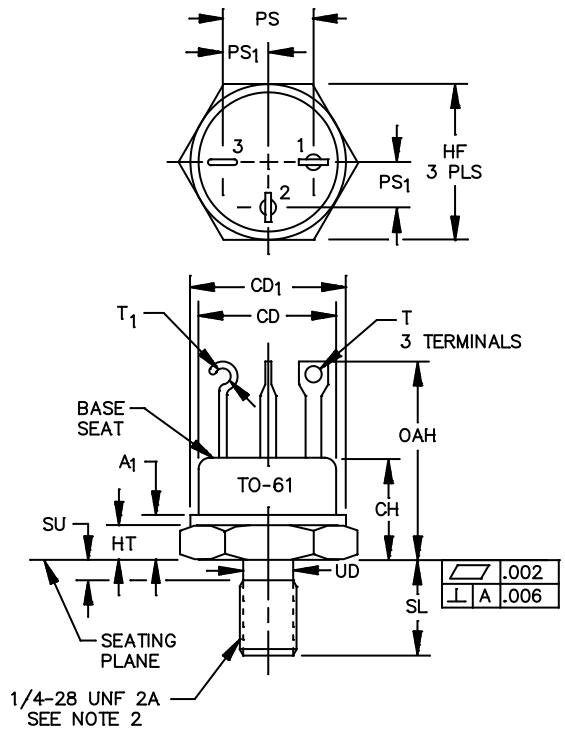
3.5 Marking. Marking shall be in accordance with MIL–PRF–19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be table I as specified herein.

3.8 Workmanship. Devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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Ltr	Dimension				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A_1		.270		6.86	
CD	.570	.610	14.48	15.49	4
CD_1	.610	.687	15.49	17.45	4
CH	.325	.460	8.26	11.68	
HF	.667	.687	16.94	17.45	
HT	.090	.150	2.29	3.81	5
OAH	.640	.875	16.26	22.22	6
PS	.340	.415	8.64	10.54	7
PS_1	.170	.213	4.32	5.41	7
SL	.422	.455	10.72	11.56	
SU		.090		2.29	8
T	.047	.072	1.19	1.83	9
T_1	.046	.077	1.17	1.96	10
UD	.220	.249	5.59	6.32	

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. See NSB Handbook H28, "Screw-Thread Standards for Federal Services". Maximum recommended mounting torque 20 in-lb (2.26 N-m).
3. Pin out: Terminal 1 = emitter, terminal 2 = base, terminal 3 = collector. The collector shall be electrically connected to the case.
4. Package contour with the exception of the hexagon is optional with dimensions specified.
5. Chamfer or undercut on one or both ends of hexagon portion is optional.
6. All three terminals.
7. Terminal spacing measured at seat only. Position of terminals in relation to hexagon is not controlled.
8. Length of incomplete or undercut threads.
9. Two leads (base and emitter).
10. This terminal can be flattened and pierced or hook type.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions TO–210AC (similar to TO–61) package.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL–PRF–19500 and as specified herein.

4.2.1 Group E qualification. Group E qualification shall be performed for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 Transient thermal impedance. The transient thermal impedance measurements shall be performed in accordance with test method 3131 of MIL–STD–750 using the guidelines in that test method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). See table II, subgroup 4 herein.

4.3 Screening (quality levels TX and TXV only). Screening shall be in accordance with table E–IV of MIL–PRF–19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement
	Quality levels TX and TXV
2	Required, T_A = +200°C
9	I_{CBO}
11	I_{EBO}, I_{CBO2} and h_{FE2}
12	See 4.3.1; 168 hours minimum
13	Subgroups 2 of table I herein; ΔI_{EBO} = 100 percent of initial value or 10 nA dc, whichever is greater; ΔI_{CBO2} = 100 percent of initial value or 10 nA dc, whichever is greater; Δh_{FE2} = ± 15 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CE} = 10 to 30 V dc; power shall be applied to achieve T_J = +175°C minimum using a minimum power dissipation equal to 75 percent of maximum rated P_T (see 1.3). NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL–PRF–19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL–PRF–19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E–VIB of MIL–PRF–19500 for JAN, JANTX, and JANTXV group B testing. Delta measurements shall be in accordance with 4.6 herein.

Subgroup	Method	Conditions
B3	1037	V_{CB} ≥ 10 V dc, 2,000 cycles.
B6	1032	T_A = +200°C.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E–VII of MIL–PRF–19500. Delta measurements shall be in accordance with 4.6 herein.

Subgroup	Method	Conditions
C1	1001	Normal mounting pressure = 8 mm Hg for 60 seconds.
C2	2036	Tension, condition A, weight = 10 lbs (4.54 Kg), time = 10 seconds.
C2	2036	Terminal torque, condition D1, torque = 6 in-oz (4.24 N-cm), time = 15 seconds.
C2	2036	Stud torque condition D2, torque = 20 in-lbs (2.26 N-m), time = 15 seconds.
C6	1037	V_{CB} ≥ 10 V dc, 6,000 cycles.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E–IX of MIL–PRF–19500 and as specified herein. Delta measurements shall be in accordance with table II herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL–STD–750.

4.6 Delta measurements. Delta measurements for groups B, C and E shall be taken as specified below.

Step	Inspection	MIL–STD–750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1	Forward current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 5.0 A dc, pulsed (see 4.5.1)	Δh_{FE2}	+20, -10 percent of initial value		

The following details shall apply:

a. The measurements for group B quality levels JAN, JANTX and JANTXV (table E–VIB of MIL–PRF–19500) are as follows: For subgroups 3 and 6, step 1.
b. The measurements for group C (table E–VII of MIL–PRF–19500) are as follows: For subgroup 6, step 1.
c. The measurements for group E (table E–IX of MIL–PRF–19500) are as follows: For subgroups 1 and 2, step 1.

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TABLE I. Group A inspection.

Inspection 1/	MIL–STD–750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3131	See 4.2.2	$Z_{\theta JX}$		7.0	°C/W
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	BV_{CEO}			
2N2812				60		V dc
2N2814				80		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 100 mA dc, pulsed (see 4.5.1)	BV_{CEO}			
2N2812				60		V dc
2N2814				80		V dc
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 10 µA dc	BV_{CBO}			
2N2812				80		V dc
2N2814				120		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 µA dc	BV_{EBO}	8.0		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 50 V dc	I_{CEO}		10	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{EB} = 0.5 V dc	I_{CEX1}			
2N2812		V_{CE} = 80 V dc			1.0	µA dc
2N2814		V_{CE} = 120 V dc			1.0	µA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
2N2812		V_{CB} = 60 V dc			0.1	µA dc
2N2814		V_{CB} = 80 V dc			0.1	µA dc
Emitter to base, cutoff current	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO}		0.1	µA dc
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE1}	50	150	
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 5.0 A dc; pulsed (see 4.5.1)	h_{FE2}	40	120	
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE3}	15		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL–STD–750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - continued						
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE4}	10		
Base to emitter voltage (non-saturated)	3066	Test condition B; V_{CE} = 5.0 V dc, I_C = 5.0 A dc, pulsed (see 4.5.1)	V_{BE}		1.2	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 5.0 A dc, I_B = 0.5 A dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.2	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 10 A dc, I_B = 1.0 A dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.0	V dc
Collector to emitter voltage (saturated)	3071	I_C = 5.0 A dc, I_B = 0.5 A dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.5	V dc
Collector to emitter voltage (saturated)	3071	I_C = 10 A dc, I_B = 1.0 A dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.5	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{EB} = 0.5 V dc	I_{CEX2}			
2N2812		V_{CE} = 60 V dc			50	µA dc
2N2814		V_{CE} = 80 V dc			50	µA dc
Low temperature operation		T_A = –55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 5.0 A dc; pulsed (see 4.5.1)	h_{FE5}	20		
Subgroup 4						
Magnitude of small-signal short- circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1.0 A dc; f = 10 MHz	$\lvert h_{fe} \rvert$	1.5	7.0	
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 5.0 V dc; I_C = 50 mA dc f = 1 kHz	h_{fe}	40	150	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		350	pF
Pulse response	3251	See figure 2				
Turn-on time		V_{CC} = 25.5 V dc; V_{BB} = 5.0 V dc; I_C = 5.0 A dc; I_{B1} = 0.5 A dc	t_{ON}		400	ns

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL–STD–750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - continued						
Storage time		V_{CC} = 25.5 V dc; V_{BB} = 5.0 V dc; I_C = 5.0 A dc; I_{B1} = I_{B2} = 0.5 A dc	t_S		1,000	ns
Fall time		V_{CC} = 25.5 V dc; V_{BB} = 5.0 V dc; I_C = 5.0 A dc; I_{B1} = I_{B2} = 0.5 A dc	t_F		400	ns
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +100°C; 1 cycle; (see figure 3); t = 1 s				
Test 1		V_{CE} = 5.0 V dc; I_C = 10 A dc				
Test 2		V_{CE} = 10 V dc; I_C = 5.0 A dc				
Test 3						
2N2812 2N2814		V_{CE} = 50 V dc; I_C = 0.2 A dc V_{CE} = 70 V dc; I_C = 0.1 A dc				
Safe operating area (switching)	3053	Load cond. C, T_C = +25°C; duty cycle ≤ 2 percent. (see figure 4)				
Test 1		t_p = 7.82 μs (vary to obtain I_C); tr = tf ≤ 500 ns; R_{BB1} = 5 Ω; V_{BB1} = 13 V dc; R_{BB2} = ∞, V_{BB2} = 0; V_{CC} = 22.5 V dc, I_C = 8 A dc, L = 15 μH, ≤ 1 Ω				
Test 2		t_p = 25 μs (vary to obtain I_C); tr = tf ≤ 500 ns; R_{BB1} = 5 Ω; V_{BB1} = 13 V dc; R_{BB2} = ∞, V_{BB2} = 0; V_{CC} = 22.5 V dc, I_C = 2.54 A dc, L = 150 μH, ≤ 10 Ω				
Electrical end-points		Table I, subgroup 2				

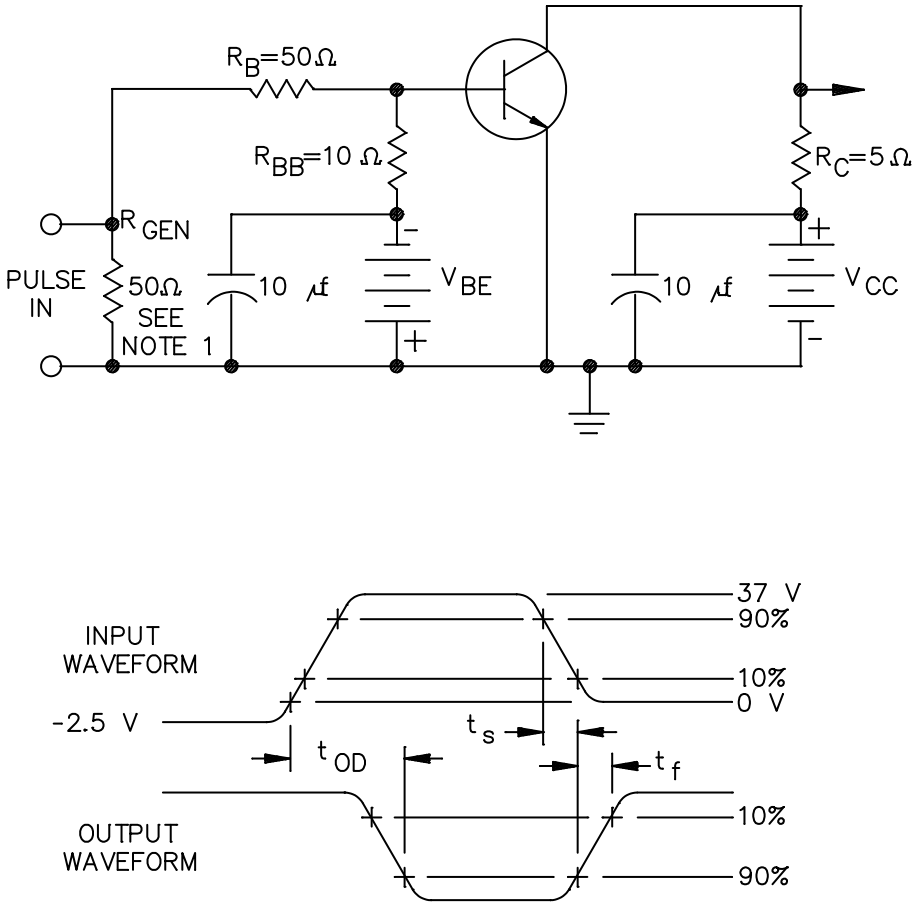
1/ For sampling plan, see MIL–PRF–19500.

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TABLE II. Group E inspection (all quality levels) - for qualification or requalification only.

Inspection	MIL–STD–750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 and 4.6 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} ≥ 10 V dc , 6,000 cycles.	
Electrical measurements		See table I, subgroup 2 and 4.6 herein.	
Subgroup 3			3 devices, c = 0
DPA	2102		
Subgroup 4			Sample size N/A
Thermal impedance curves		Each supplier shall submit their (typical) design thermal impedance curves. In addition, test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report.	
Subgroup 5 and 6			
Not applicable			
Subgroup 7			45 devices c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V dc. Condition B for devices ≤ 400 V dc.	

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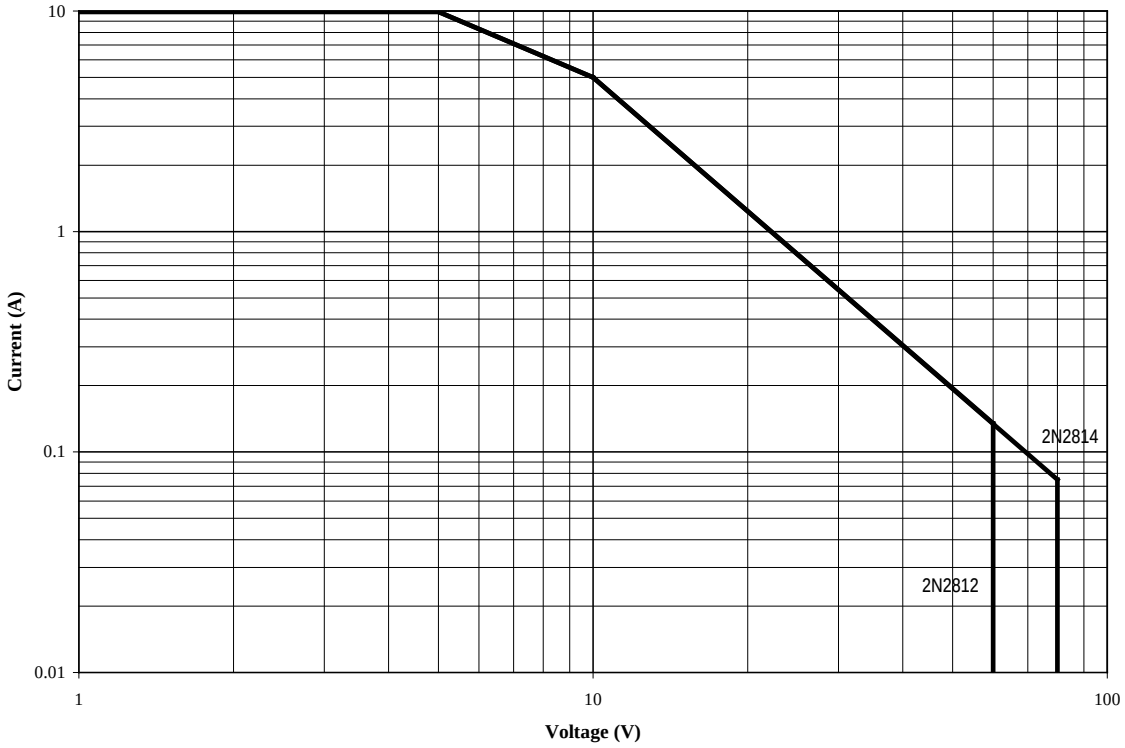


NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: pulse width = 20 μs, Z_{out} = 50Ω, pulse repetition rate = 1 kHz, rise time (t_r) and fall time (t_f) ≤ 10 ns, duty cycle ≤ 2 percent.
2. The output waveform is monitored on a sampling oscilloscope with t_f ≤ 15 ns, R_{in} ≥ 10 MΩ and C_{IN} ≤ 11.5 pF.

FIGURE 2. Pulse response test circuit.

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NOTE: T_C = +100°C

FIGURE 3. Maximum safe operating area (forward bias, continuous DC).

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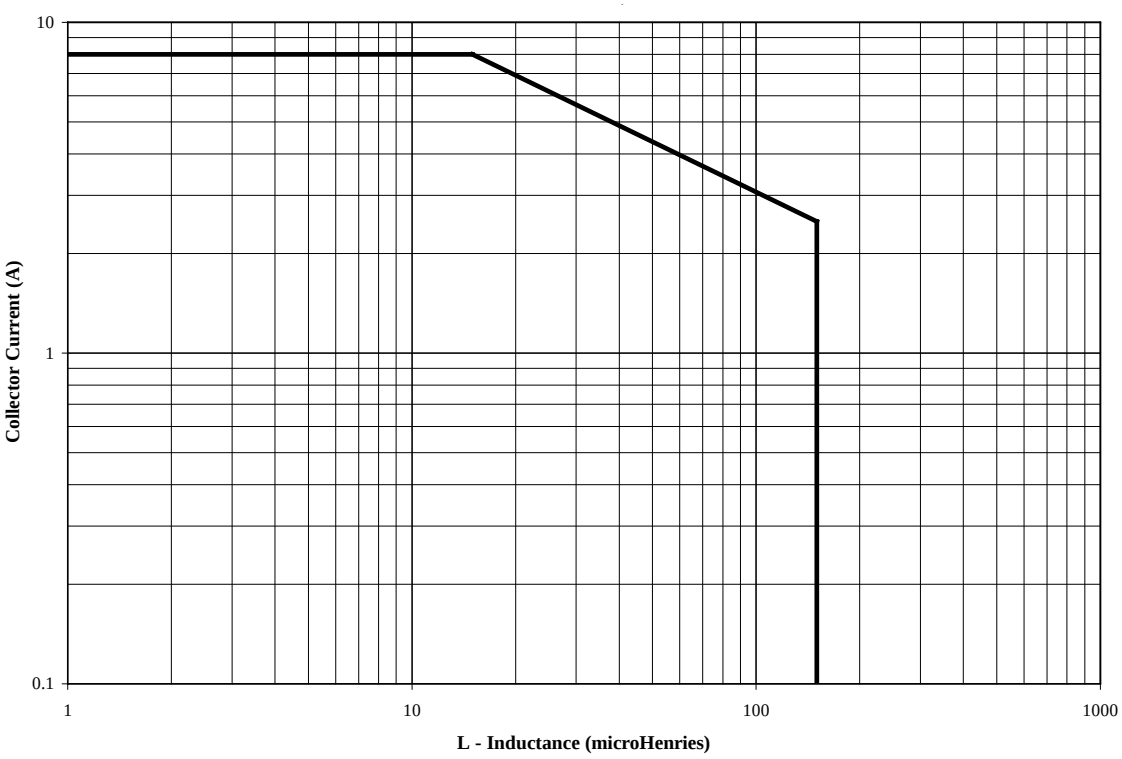


FIGURE 4. Maximum safe operating area (switching unclamped inductive load).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL–PRF–19500 are applicable to this specification.)

6.1 Intended use. Transistors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

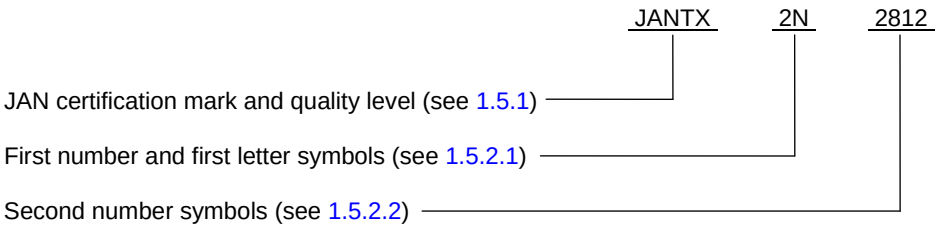
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land ad Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are in the following form.



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6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level
JAN2N2812	JANTX2N2812	JANTXV2N2812
JAN2N2814	JANTX2N2814	JANTXV2N2814

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR
Air Force – 85
DLA – CC

Review activities:
Army – MI
Air Force – 19, 70, 99

Preparing activity:
DLA – CC

(Project 5961–2014–005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:45Z
Check the source to verify that this is the current version before use.